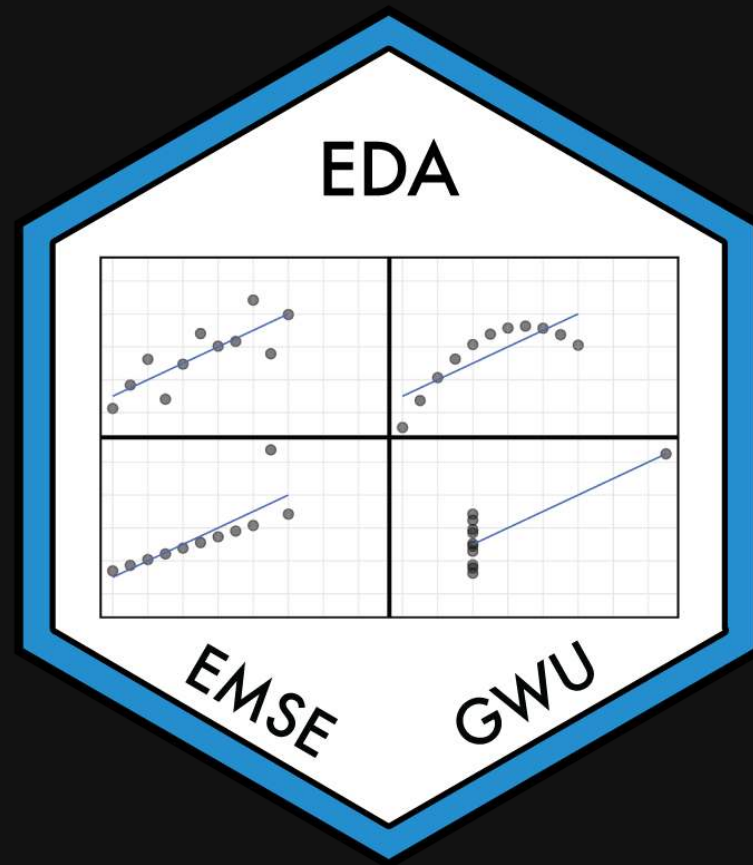




Getting Started

🏛️ EMSE 4572/6572: Exploratory Data Analysis

👤 John Paul Helveston

📅 August 26, 2026

Week 1: *Getting Started*

1. Course Goal
2. Course Introduction
3. Quarto
4. Workflow & Reading In Data
5. Wrangling Data
6. Visualizing Data

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Course 1: Intro to Programming for Analytics

"Computational Literacy"

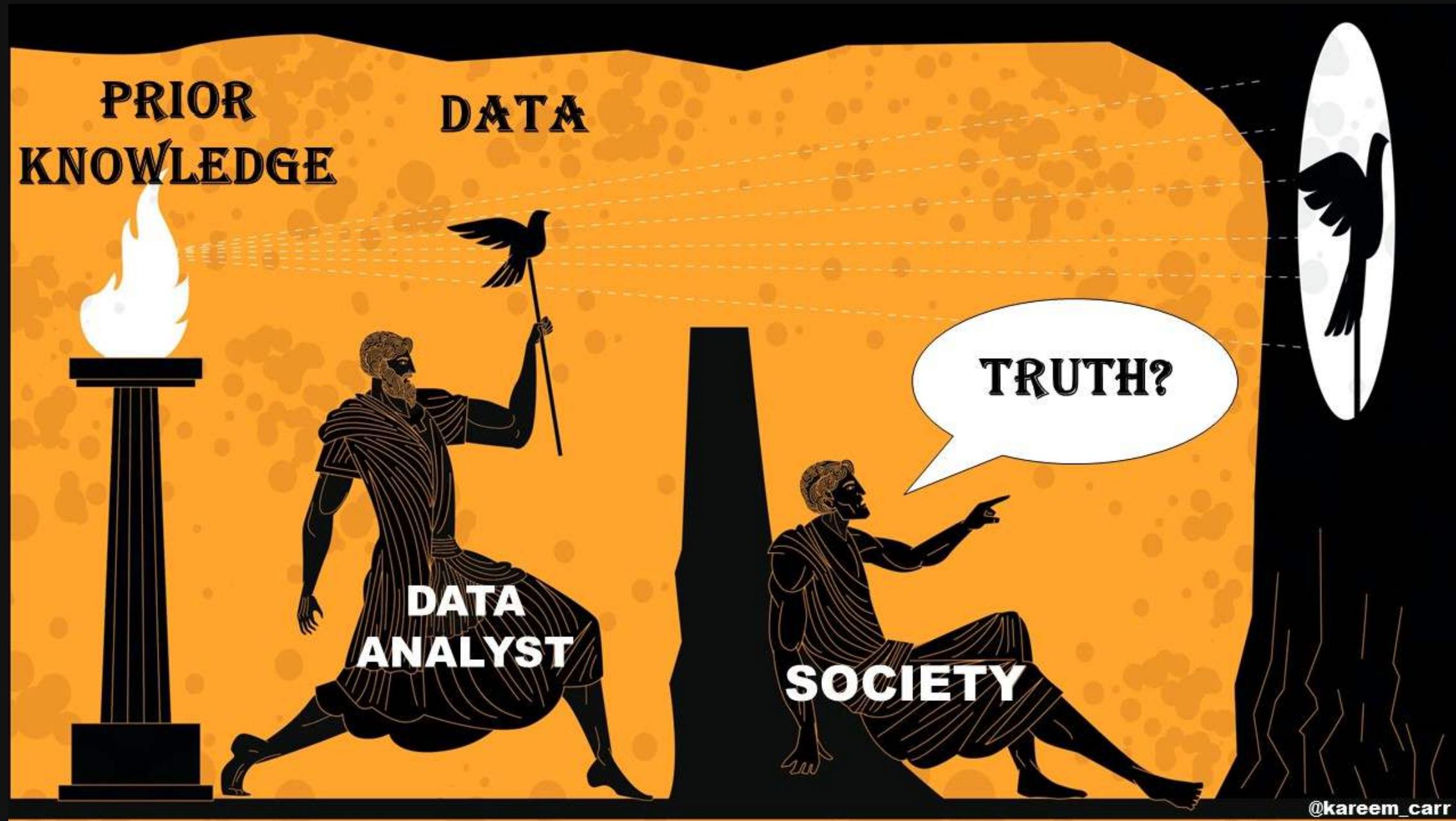
- Programming: Conditionals (if/else), loops, functions, testing, data types.
- Analytics: Data structures, import / export, basic data manipulation & visualization.

Course 2: Exploratory Data Analysis

"Data Literacy"

- Strategies for conducting an exploratory data analysis.
- Design principles for visualizing and communicating *information* extracted from data.
- Reproducibility: Reports that contain code, equations, visualizations, and narrative text.

Class goal: translate *data* into *information*



Class goal: translate *data* into *information*

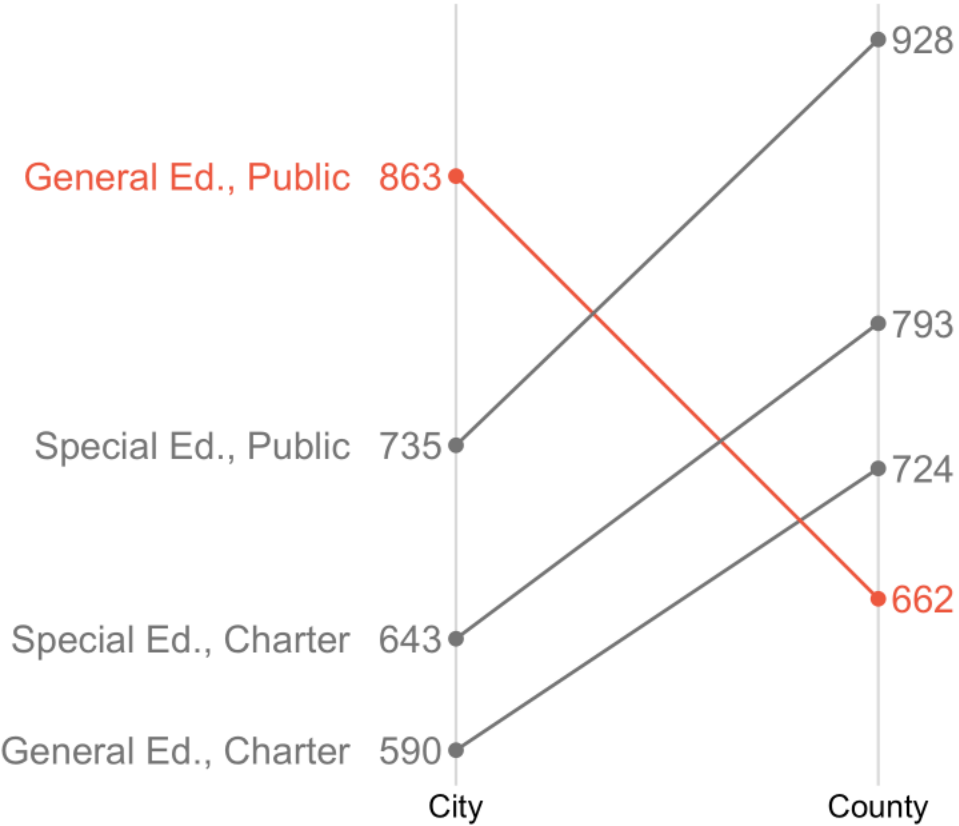
Data

Average student engagement scores

Class	Type	City	County
Special Ed.	Charter	643	793
Special Ed.	Public	735	928
General Ed.	Charter	590	724
General Ed.	Public	863	662

Information

Students in public, general education classes in county schools have surprisingly low engagement



Data exploration: an iterative process

Encode data:

```
1 engagement_data <- data.frame(  
2   City = c(643, 735, 590, 863),  
3   County = c(793, 928, 724, 662),  
4   School = c(  
5     "Special Ed., Charter",  
6     "Special Ed., Public",  
7     "General Ed., Charter",  
8     "General Ed., Public"  
9   )  
10 )  
11  
12 engagement_data
```

```
#>   City County      School  
#> 1  643    793 Special Ed., Charter  
#> 2  735    928 Special Ed., Public  
#> 3  590    724 General Ed., Charter  
#> 4  863    662 General Ed., Public
```

Re-format data for plotting:

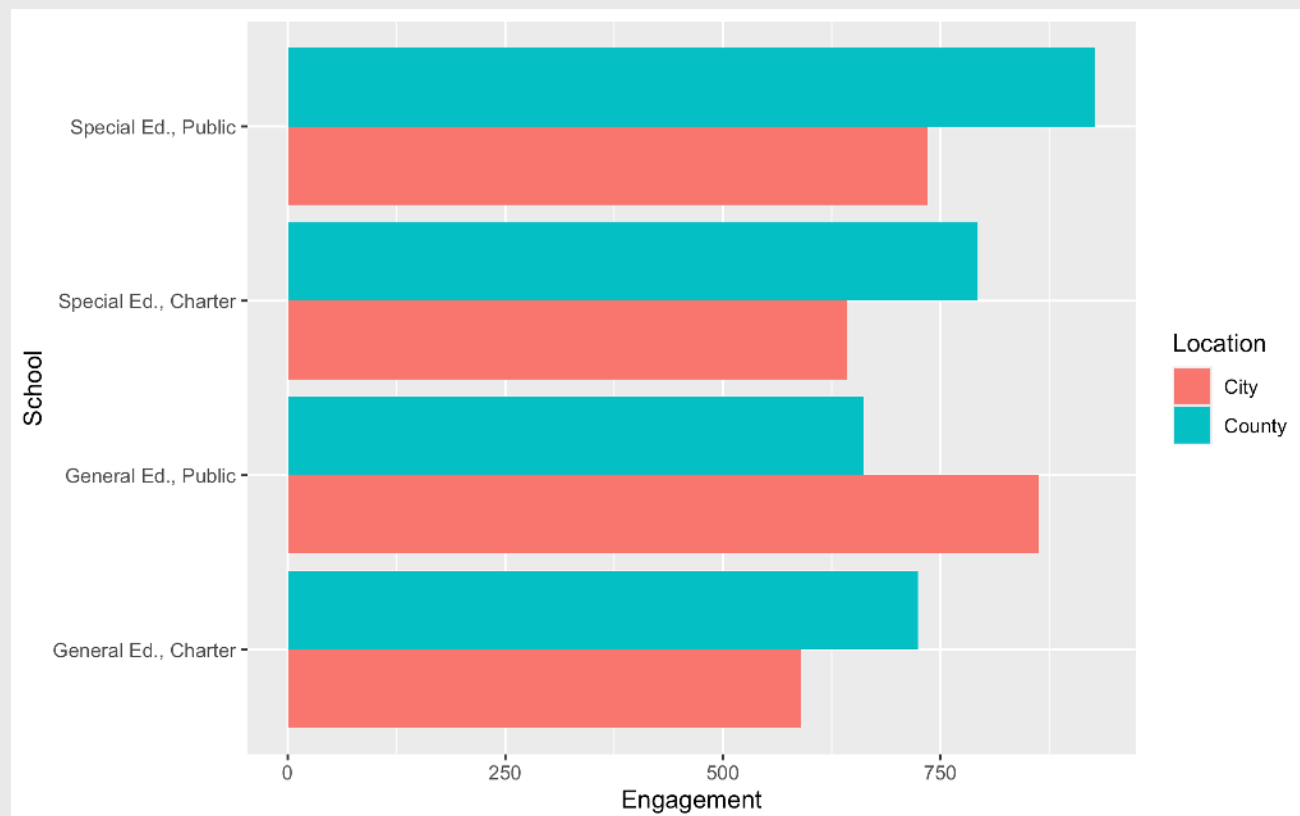
```
1 engagement_data <- engagement_data %>%  
2   pivot_longer(  
3     cols = c(City, County),  
4     names_to = "Location",  
5     values_to = "Engagement"  
6   )  
7  
8 engagement_data
```

```
#> # A tibble: 8 × 3  
#>   School      Location Engagement  
#>   <chr>      <chr>      <dbl>  
#> 1 Special Ed., Charter City         643  
#> 2 Special Ed., Charter County        793  
#> 3 Special Ed., Public City         735  
#> 4 Special Ed., Public County        928  
#> 5 General Ed., Charter City         590  
#> 6 General Ed., Charter County        724  
#> 7 General Ed., Public City         863  
#> 8 General Ed., Public County        662
```

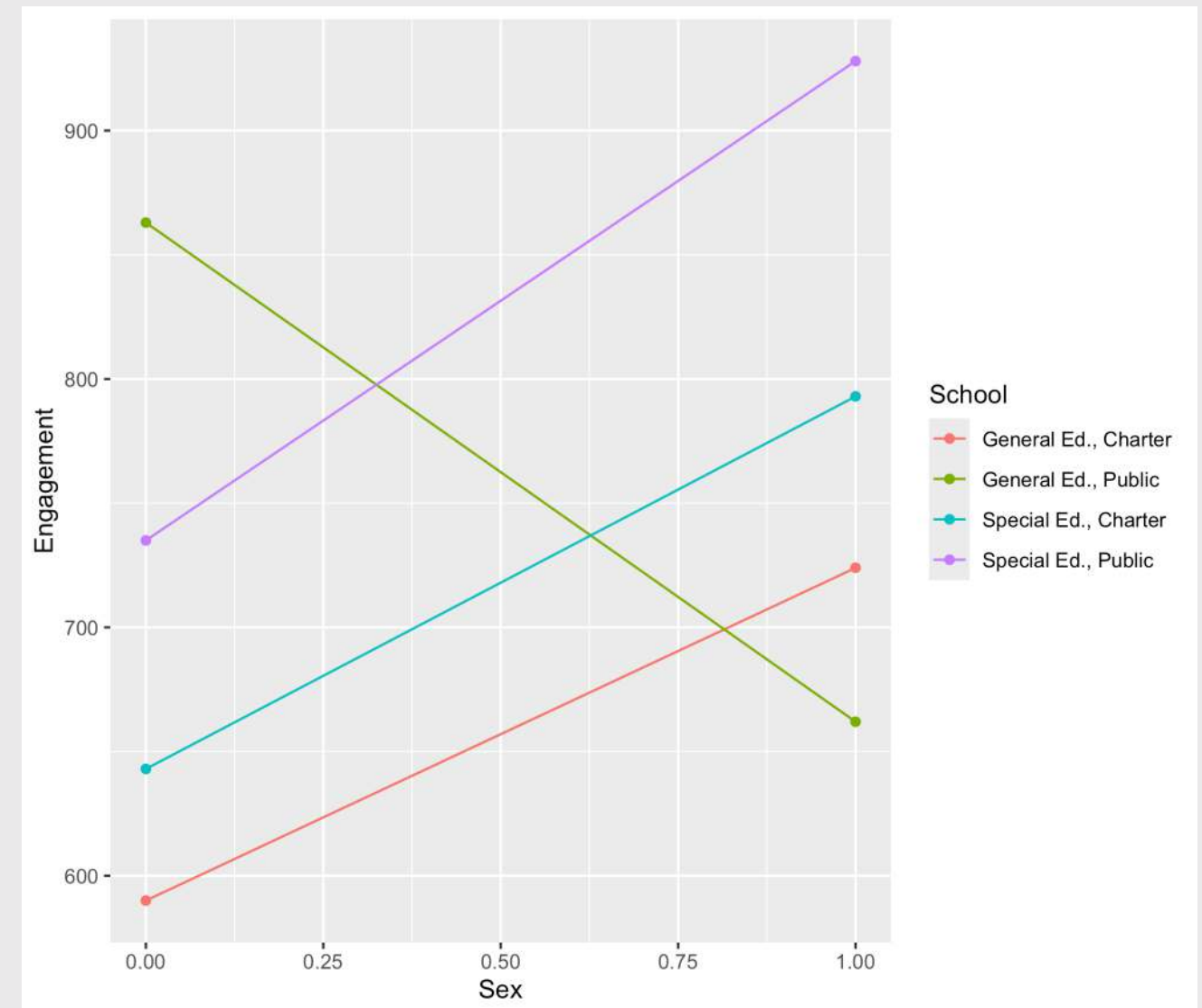

Data exploration: an iterative process

Initial exploratory plotting:

```
1 engagement_data %>%  
2   ggplot() +  
3   geom_col(  
4     aes(x = Engagement, y = School, fill = Location),  
5     position = "dodge"  
6   )
```

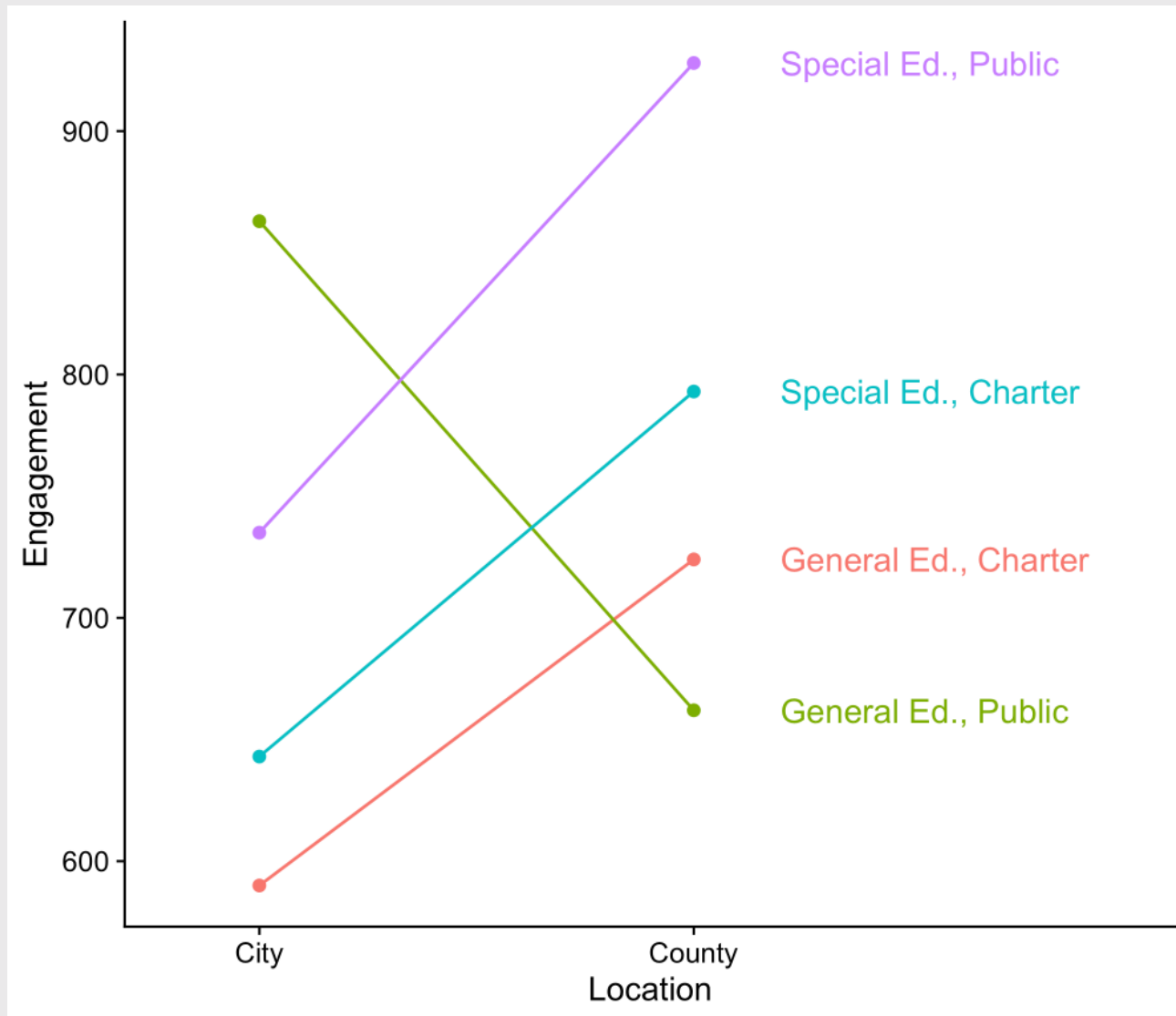


More exploratory plotting:
highlight difference

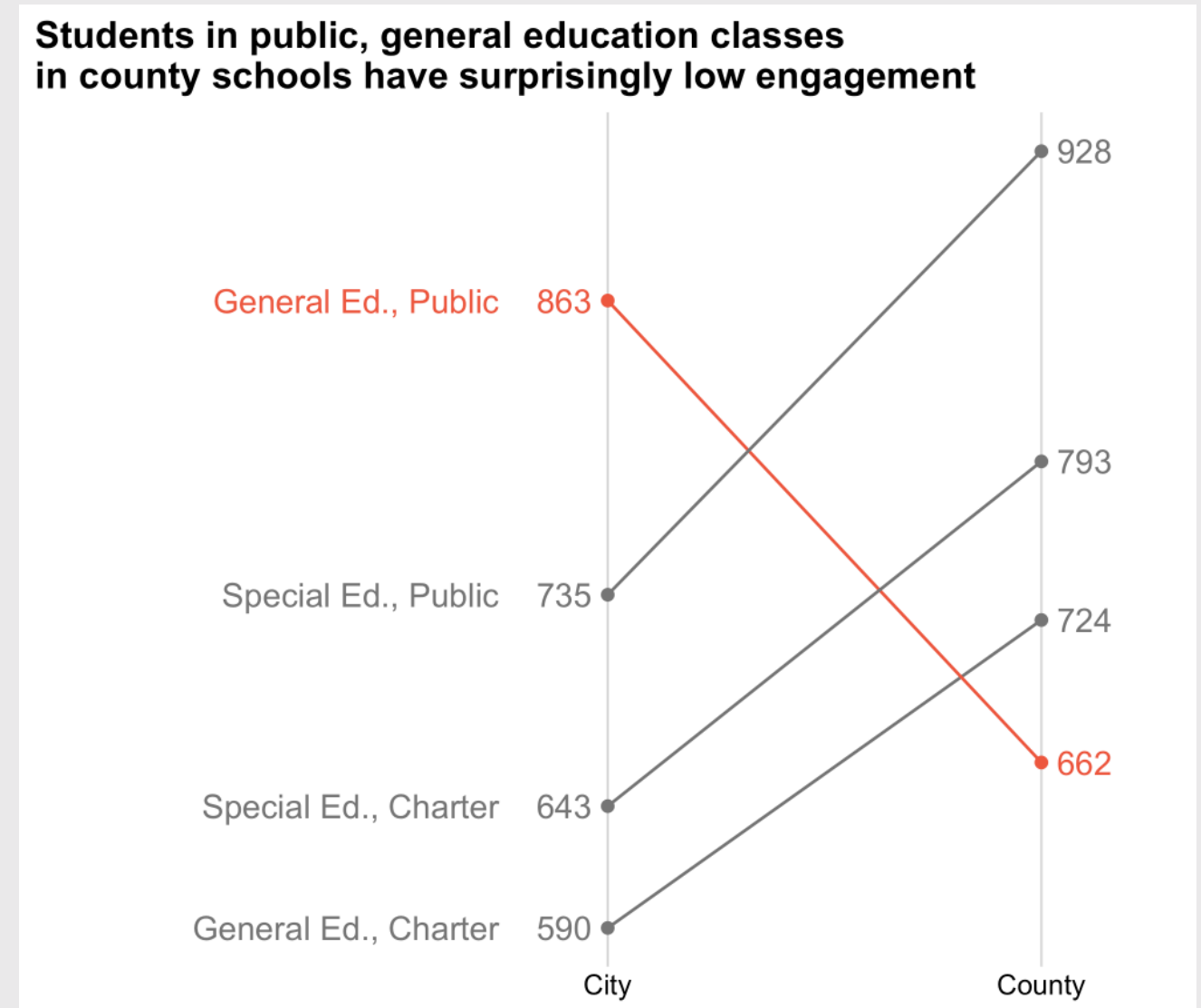


Data exploration: an iterative process

Directly label figure:



Remove unnecessary axes, change colors, fix labels:



A fully reproducible analysis

Code Plot

Clean the data

```
1 data <- data.frame(  
2   City = c(643, 735, 590, 863),  
3   County = c(793, 928, 724, 662),  
4   School = c(  
5     "Special Ed., Charter", "Special Ed., Public",  
6     "General Ed., Charter", "General Ed., Public"  
7   ),  
8   Highlight = c(0, 0, 0, 1)  
9 ) %>%  
10 pivot_longer(  
11   cols = c(City, County),  
12   names_to = "Location",  
13   values_to = "Engagement"  
14 ) %>%  
15 mutate(  
16   Location = fct_relevel(Location, c("City", "County")),  
17   Highlight = as.factor(Highlight),  
18   x = ifelse(Location == "County", 1, 0)  
19 )
```

Make the plot

```
1 plot <- ggplot(data, aes(  
2   x = x, y = Engagement, group = School, color = Highlight  
3 )) +  
4   geom_point() +  
5   geom_line() +  
6   scale_color_manual(values = c("#757575", "#ed573e")) +  
7   labs(  
8     x = "Sex", y = "Engagement",  
9     title = "County general-ed classes have low engagement"  
10  ) +  
11   scale_x_continuous(  
12     limits = c(-1.2, 1.2),  
13     labels = c("City", "County"), breaks = c(0, 1)  
14  ) +  
15   geom_text_repel(aes(label = Engagement), size = 5) +  
16   theme_cowplot() +  
17   background_grid(major = "x") +  
18   theme(legend.position = "none")
```

Data exploration: an iterative process

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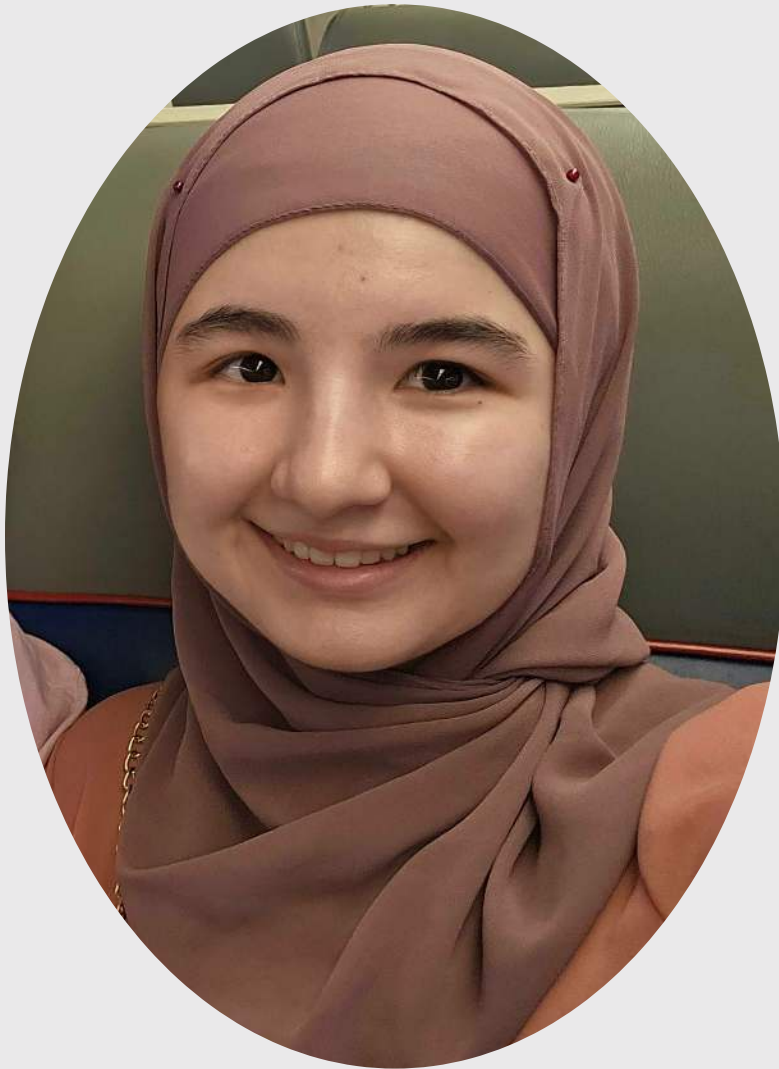
Meet your instructor!



John Helveston, Ph.D.

- 2025 - Present: Associate Professor, EMSE (w/tenure)
- 2018 - 2025: Assistant Professor, EMSE
- 2016-2018: Postdoc at [Institute for Sustainable Energy](#), Boston University
- 2016: PhD in Engineering & Public Policy at Carnegie Mellon University
- 2015: MS in Engineering & Public Policy at Carnegie Mellon University
- 2010: BS in Engineering Science & Mechanics at Virginia Tech
- Website: www.jhelvy.com

Meet your tutor!



Lola Nurullaeva

- Graduate Teaching Assistant (GTA)
- EMSE Ph.D. Student (B.S. SE & M.S. Data Analytics)
- Check out her team's [project](#) from 2023

Prerequisites

EMSE 4574 / 6574: Intro to Programming for Analytics

You should be able to:

- Use Positron to write basic R commands.
- Know the distinctions between different R operators and data types, including numeric, string, and logical data.
- Use **tidyverse** functions to wrangle and manipulate data in R.
- Use the **ggplot2** library to create plots in R.

 [Check out R for Analytics Primer](#)

Course website

 Everything you need will be on the course website:
<https://eda.seas.gwu.edu/2026-Fall/>

 The [schedule](#) is the best starting point

Quizzes (10% of grade)

 At the start of class every other week-ish. Make ups only for excused absences (i.e. don't be late).

 5 total, lowest dropped

 10 minutes

Why quiz at all? The “retrieval effect” - basically, you have to *practice* remembering things, otherwise your brain won't remember them (see the book [“Make It Stick: The Science of Successful Learning”](#))

Assignments

- 1)  Weekly Reflections: [HW1](#)
- 2)  3 Mini Projects (due 2 weeks from date assigned)
- 3)  [Final Project](#)

Undergrads: Teams of 3 - 4 students

Grads: Teams of 2 students

Item	Due Date
Proposal	Sep 20
Initial Report	Nov 01
Final Report	Dec 06
Presentation	Dec 08

Grades

Item	Weight	Notes
Participation / Attendance	5 %	(Yes, I take attendance)
Reflections	13 %	Weekly assignment, lowest dropped
Quizzes	10 %	5 quizzes, lowest dropped
Mini Project 1	9 %	Individual assignments
Mini Project 2	9 %	
Mini Project 3	9 %	
Final Project: Proposal	6 %	
Final Project: Progress Report	6 %	Team project
Final Project: Report	17 %	
Final Project: Presentation	6 %	
Final Interview	10 %	Individual interview

Grades

Course policies

BE NICE

BE HONEST

DON'T CHEAT

Copying is good, stealing is bad

“Plagiarism is trying to pass someone else’s work off as your own. Copying is about reverse-engineering.”

– Austin Kleon, from [Steal Like An Artist](#)

Use of AI

We will heavily use AI tools this semester

- Large language models (LLMs) are pretty good
- Sometimes they suck.
- I will grade whatever you submit. **It should not suck.**

Ways to not have your work suck:

- Don't submit code that doesn't run (actually run it before submitting it).
- Actually read what the AI generates, and **don't submit something you don't understand**. (Ask the LLM to explain *why* something works or doesn't work)

Use the approach I teach: agentic workflows (starting next week)

Late submissions

- **3** late days - use them anytime, no questions asked
- No more than **2** late days on any one assignment
- Contact me for special cases

How to succeed in this class

 Participate during class!

 Start assignments early and **read carefully!**

 Actually read (before class)!

 Get sleep and take breaks often!

 Ask for help!

Getting Help

🔗 Use [Slack](#) to ask questions.

👤 Meet with your tutors

🕒 [Schedule a meeting](#) w/Prof. Helveston (Mon/Tue/Fri)

</> [GW Coders](#)

Course Software

 **Slack: turn notifications on!**

 **R & Positron** (Install both)

 **Posit Cloud** (Register for free!)

 **Quarto**

 **GitHub Desktop**

Break

1. Install everything on the [software page](#)
2. Add your **GitHub username** (NOT GW netID) in [this sheet](#) next to your name

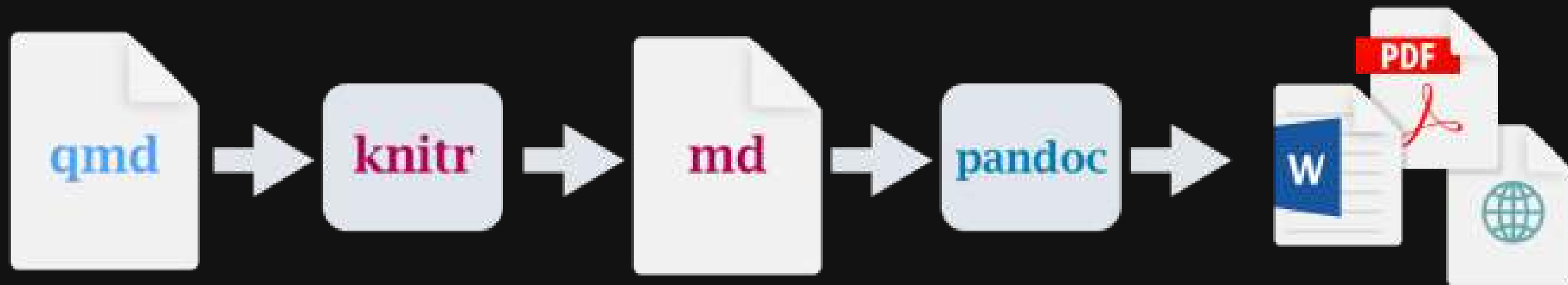
05 : 00

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Quick demo

1. Open `quarto_demo.qmd`
2. Click “Preview”



Anatomy of a .qmd file

Header

Markdown text

R code

Define overall document options in header

Basic html page

```
1 ---
2 title: Your title
3 author: Author name
4 format: html
5 ---
```

Add table of contents, change theme

```
1 ---
2 title: Your title
3 author: Author name
4 toc: true
5 format:
6   html:
7     theme: united
8 ---
```

More on themes at

<https://quarto.org/docs/output-formats/html-themes.html>

Render to multiple outputs

PDF uses LaTeX

```
1 ----  
2 title: Your title  
3 author: Author name  
4 format: pdf  
5 ----
```

Microsoft Word

```
1 ----  
2 title: Your title  
3 author: Author name  
4 format: docx  
5 ----
```

If you don't have LaTeX on your computer,
install tinytex in R:

```
1 tinytex::install_tinytex()
```

Anatomy of a .qmd file

~~Header~~

Markdown text

R code

Right now, bookmark this! 🙌

<https://commonmark.org/help/>

(When you have 10 minutes, do this! 🙌)

<https://commonmark.org/help/tutorial/>

Headers

```
1 # HEADER 1
2
3 ## HEADER 2
4
5 ### HEADER 3
6
7 #### HEADER 4
8
9 ##### HEADER 5
10
11 ##### HEADER 6
```

HEADER 1

HEADER 2

HEADER 3

HEADER 4

HEADER 5

HEADER 6

Basic Text Formatting

Type this...

- `normal text`
- `_italic text_`
- `*italic text*`
- `**bold text**`
- `***bold italic text***`
- `~~strikethrough~~`
- ``code text``

..to get this

- normal text
- *italic text*
- *italic text*
- **bold text**
- ***bold italic text***
- ~~strikethrough~~
- `code text`

Lists

Bullet list:

```
1 - first item  
2 - second item  
3 - third item
```

- first item
- second item
- third item

Numbered list:

```
1 1. first item  
2 2. second item  
3 3. third item
```

1. first item
2. second item
3. third item

Links

Simple **url link** to another site:

```
1 [Download R](http://www.r-project.org/)
```

[Download R](http://www.r-project.org/)

Anatomy of a .qmd file

~~Header (think of this as the “settings”)~~

~~Markdown text~~

R code

R Code

Inline code

```
1 `r insert code here`
```

Code chunks

```
1 ```{r}  
2 insert code here  
3 insert more code here  
4 ```
```

Inline R code

```
1 The sum of 3 and 4 is `r 3 + 4`
```

Produces this:

The sum of 3 and 4 is 7

R Code chunks

This code chunk...

```
1 ```{r}
2 library(palmerpenguins)
3
4 glimpse(penguins)
5 ```
```

...will produce this when rendered:

```
1 library(palmerpenguins)
2
3 glimpse(penguins)
```

```
#> Rows: 4
#> Columns: 8
#> $ species      <fct> Adelie, Adelie, Adelie, Adelie
#> $ island       <fct> Torgersen, Torgersen, Torgersen,
Torgersen
#> $ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA
#> $ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA
#> $ flipper_length_mm <int> 181, 186, 195, NA
#> $ body_mass_g    <int> 3750, 3800, 3250, NA
#> $ sex           <fct> male, female, female, NA
#> $ year          <int> 2007, 2007, 2007, 2007
```

Chunk options

Control what chunks output using options

All options [here](#)

option	default	effect
eval	TRUE	Whether to evaluate the code and include its results
echo	TRUE	Whether to display code along with its results
warning	TRUE	Whether to display warnings
error	FALSE	Whether to display errors
message	TRUE	Whether to display messages
tidy	FALSE	Whether to reformat code in a tidy way when displaying it
results	"markup"	"markup", "asis", "hold", or "hide"
cache	FALSE	Whether to cache results for future renders
comment	"##"	Comment character to preface results with
fig.width	7	Width in inches for plots created in chunk
fig.height	7	Height in inches for plots created in chunk

Chunk output options

By default, code chunks print **code** + **output**

```
1 ```{r}
2 #| echo: false
3
4 cat('hello world!')
5 ```
```

Prints only **output**
(doesn't show code)

```
#> hello world!
```

```
1 ```{r}
2 #| eval: false
3
4 cat('hello world!')
5 ```
```

Prints only **code**
(doesn't run the code)

```
1 cat("hello world!")
```

```
1 ```{r}
2 #| include: false
3
4 cat('hello world!')
5 ```
```

Runs, but doesn't print
anything

A global `setup` chunk

```
1  ```{r}
2  #| label: setup
3  #| include: false
4
5  knitr::opts_chunk$set(
6    warning = FALSE,
7    message = FALSE,
8    fig.width = 7.252,
9    fig.height = 4,
10   comment = "#>"
11 )
12  ```
```

- Typically the first chunk
- All following chunks will use these options (i.e., sets global chunk options)
- You can (and should) use individual chunk options too
- Often where I load libraries, etc.

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Workflow for reading in data

1. Open your project **folder** in Positron - **don't double-click .R files!**
2. Build file paths with `file.path()` — relative to the folder you opened

```
1 path <- file.path("folder", "file.csv")
```

3. Import data with these functions:

File type	Function	Library
.csv	<code>read_csv()</code>	readr
.txt	<code>read.table()</code>	utils
.xlsx	<code>read_excel()</code>	readxl

Importing Comma Separated Values (.csv)

Read in `.csv` files with `read_csv()`:

```
1 library(tidyverse)
2
3 csvPath <- file.path("data", "milk_production.csv")
4 milk_production <- read_csv(csvPath)
5
6 head(milk_production)
```

```
#> # A tibble: 6 × 4
#>   region      state      year milk_produced
#>   <chr>      <chr>      <dbl>      <dbl>
#> 1 Northeast Maine      1970      619000000
#> 2 Northeast New Hampshire 1970      356000000
#> 3 Northeast Vermont    1970     1970000000
#> 4 Northeast Massachusetts 1970      658000000
```

Importing Text Files (.txt)

Read in `.txt` files with `read.table()`:

```
1 txtPath <- file.path("data", "nasa_global_temps.txt")
2 global_temps <- read.table(txtPath, skip = 5, header = FALSE)
3
4 head(global_temps)
```

```
#>      V1      V2      V3
#> 1 1880 -0.15 -0.08
#> 2 1881 -0.07 -0.12
#> 3 1882 -0.10 -0.15
#> 4 1883 -0.16 -0.19
#> 5 1884 -0.27 -0.23
#> 6 1885 -0.32 -0.25
```

Importing Text Files (.txt)

Read in `.txt` files with `read.table()`:

```
1 txtPath <- file.path("data", "nasa_global_temps.txt")
2 global_temps <- read.table(txtPath, skip = 5, header = FALSE)
3 names(global_temps) <- c("year", "no_smoothing", "loess") # Add header
4
5 head(global_temps)
```

```
#>   year no_smoothing loess
#> 1 1880      -0.15 -0.08
#> 2 1881      -0.07 -0.12
#> 3 1882      -0.10 -0.15
#> 4 1883      -0.16 -0.19
#> 5 1884      -0.27 -0.23
#> 6 1885      -0.32 -0.25
```

Importing Excel Files (.xlsx)

Read in `.xlsx` files with `read_excel()`:

```
1 library(readxl)
2
3 xlsxPath <- file.path("data", "pv_cell_production.xlsx")
4 pv_cells <- read_excel(xlsxPath, sheet = "Cell Prod by Country", skip = 2)
```

```
1 glimpse(pv_cells)
```

```
#> Rows: 6
#> Columns: 10
#> $ Year      <chr> NA, NA, "1995", "1996", "1997", "1998"
#> $ China     <chr> "Megawatts", NA, "NA", "NA", "NA", "NA"
#> $ Taiwan    <chr> NA, NA, "NA", "NA", "NA", "NA"
#> $ Japan     <dbl> NA, NA, 16.4, 21.2, 35.0, 49.0
#> $ Malaysia  <chr> NA, NA, "NA", "NA", "NA", "NA"
#> $ Germany   <chr> NA, NA, "NA", "NA", "NA", "NA"
#> $ `South Korea` <chr> NA, NA, "NA", "NA", "NA", "NA"
#> $ `United States` <dbl> NA, NA, 34.75, 38.85, 51.00, 53.70
#> $ Others    <chr> NA, NA, "NA", "NA", "NA", "NA"
#> $ World     <dbl> NA, NA, 77.6, 88.6, 125.8, 154.9
```

Importing Excel Files (.xlsx)

Read in `.xlsx` files with `read_excel()`:

```
1 library(readxl)
2
3 xlsxPath <- file.path("data", "pv_cell_production.xlsx")
4 pv_cells <- read_excel(xlsxPath, sheet = "Cell Prod by Country", skip = 2) %>%
5   mutate(Year = as.numeric(Year)) %>% # Convert "non-years" to NA
6   filter(!is.na(Year)) # Drop NA rows in Year
```

```
1 glimpse(pv_cells)
```

```
#> Rows: 6
#> Columns: 10
#> $ Year      <dbl> 1995, 1996, 1997, 1998, 1999, 2000
#> $ China    <chr> "NA", "NA", "NA", "NA", "NA", "2.5"
#> $ Taiwan   <chr> "NA", "NA", "NA", "NA", "NA", "NA"
#> $ Japan    <dbl> 16.4, 21.2, 35.0, 49.0, 80.0, 128.6
#> $ Malaysia <chr> "NA", "NA", "NA", "NA", "NA", "NA"
#> $ Germany  <chr> "NA", "NA", "NA", "NA", "NA", "22.5"
#> $ `South Korea` <chr> "NA", "NA", "NA", "NA", "NA", "NA"
#> $ `United States` <dbl> 34.75, 38.85, 51.00, 53.70, 60.80, 75.00
```

Your turn

Open the `practice.qmd` file.

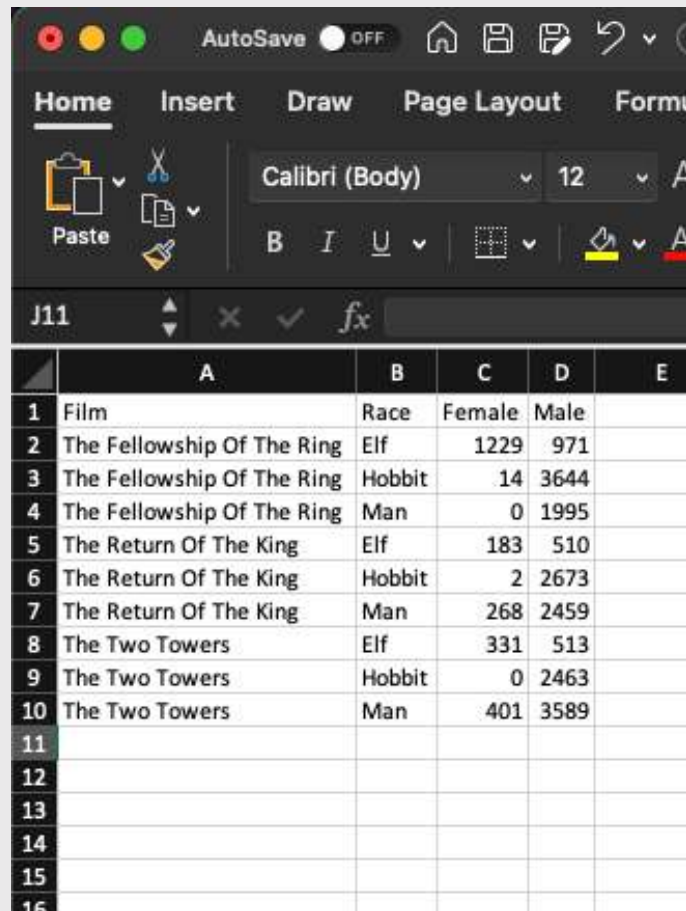
Write code to import the following data files from the "data" folder:

- For `lotr_words.csv`, call the data frame `lotr`
- For `north_america_bear_killings.txt`, call the data frame `bears`
- For `uspto_clean_energy_patents.xlsx`, call the data frame `patents`

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The data frame... in Excel



	A	B	C	D	E
1	Film	Race	Female	Male	
2	The Fellowship Of The Ring	Elf	1229	971	
3	The Fellowship Of The Ring	Hobbit	14	3644	
4	The Fellowship Of The Ring	Man	0	1995	
5	The Return Of The King	Elf	183	510	
6	The Return Of The King	Hobbit	2	2673	
7	The Return Of The King	Man	268	2459	
8	The Two Towers	Elf	331	513	
9	The Two Towers	Hobbit	0	2463	
10	The Two Towers	Man	401	3589	
11					
12					
13					
14					
15					
16					

The data frame... in R

```
1 lotr

#> # A tibble: 18 × 4
#>   film                                race gender word_count
#>   <chr>                                <chr> <chr>         <dbl>
#> 1 The Fellowship Of The Ring Elf      Female      1229
#> 2 The Fellowship Of The Ring Elf      Male        971
#> 3 The Fellowship Of The Ring Hobbit    Female        14
#> 4 The Fellowship Of The Ring Hobbit    Male      3644
#> 5 The Fellowship Of The Ring Man      Female         0
#> 6 The Fellowship Of The Ring Man      Male      1995
#> 7 The Return Of The King Elf      Female      183
#> 8 The Return Of The King Elf      Male       510
#> 9 The Return Of The King Hobbit    Female         2
#> 10 The Return Of The King Hobbit    Male      2673
#> 11 The Return Of The King Man      Female       268
#> 12 The Return Of The King Man      Male      2459
#> 13 The Two Towers Elf      Female       331
#> 14 The Two Towers Elf      Male       513
#> 15 The Two Towers Hobbit    Female         0
#> 16 The Two Towers Hobbit    Male      2463
#> 17 The Two Towers Man      Female       401
#> 18 The Two Towers Man      Male      3589
```

Columns: *Vectors* of values (must be same data type)

Extract a column using `$`

```
1 lotr$race
```

```
#> [1] "Elf"      "Elf"      "Hobbit"   "Hobbit"   "Man"      "Man"      "Elf"      "Elf"
    "Hobbit" "Hobbit"   "Man"      "Man"      "Elf"      "Elf"      "Hobbit"   "Hobbit"   "Man"
    "Man"
```

Columns: *Vectors* of values (must be same data type)

Can also use brackets:

```
1 lotr$race
```

```
#> [1] "Elf"      "Elf"      "Hobbit"   "Hobbit"   "Man"      "Man"      "Elf"      "Elf"
"Hobbit"   "Hobbit"   "Man"      "Man"      "Elf"      "Elf"      "Hobbit"   "Hobbit"   "Man"
"Man"
```

```
1 lotr[, 2]
```

```
#> # A tibble: 18 × 1
#>   race
#>   <chr>
#> 1 Elf
#> 2 Elf
#> 3 Hobbit
#> 4 Hobbit
#> 5 Man
#> 6 Man
```

Rows: Information about individual observations

Information about the first row:

```
1 lotr[1, ]
```

```
#> # A tibble: 1 × 4  
#>   film          race gender word_count  
#>   <chr>         <chr> <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Elf    Female    1229
```

Information about rows 1 & 2:

```
1 lotr[1:2, ]
```

```
#> # A tibble: 2 × 4  
#>   film          race gender word_count  
#>   <chr>         <chr> <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Elf    Female    1229  
#> 2 The Fellowship Of The Ring Elf    Male     971
```

Quick Practice

Read in the `data.csv` file in the "data" folder:

```
1 data <- read_csv(file.path("data", "data.csv"))
```

Now answer these questions:

- How many rows and columns are in the data frame?
- What type of data is each column?
- Preview the different columns - what do you think this data is about? What might one row represent?
- How many unique airlines are in the data frame?
- What is the shortest and longest air time for any one flight in the data frame?

The tidyverse: `stringr` + `dplyr` + `readr` + `ggplot2` + ...



Art by [Allison Horst](#)

The main `dplyr` “verbs”

“Verb”	What it does
<code>select()</code>	Select columns by name
<code>filter()</code>	Keep rows that match criteria
<code>arrange()</code>	Sort rows based on column(s)
<code>mutate()</code>	Create new columns
<code>summarize()</code>	Create summary values

Core `tidyverse` concept: Chain functions together with “pipes”

`%>%`

Think of the words “...and then...”

```
1 data %>%  
2   do_something() %>%  
3   do_something_else()
```


Select columns with `select()`

Subset Variables (Columns)



Select columns with `select()`

Select the columns `film` & `race`

```
1 lotr %>%  
2   select(film, race)
```

```
#> # A tibble: 18 × 2  
#>   film                race  
#>   <chr>              <chr>  
#> 1 The Fellowship Of The Ring Elf  
#> 2 The Fellowship Of The Ring Elf  
#> 3 The Fellowship Of The Ring Hobbit  
#> 4 The Fellowship Of The Ring Hobbit  
#> 5 The Fellowship Of The Ring Man  
#> 6 The Fellowship Of The Ring Man  
#> 7 The Return Of The King      Elf  
#> 8 The Return Of The King      Elf
```

Select columns with `select()`

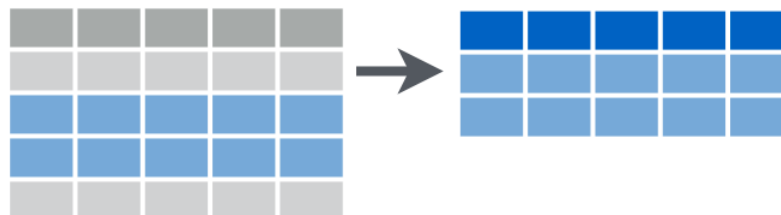
Use the `-` sign to drop columns

```
1 lotr %>%  
2   select(-film)
```

```
#> # A tibble: 18 × 3  
#>   race    gender word_count  
#>   <chr>  <chr>      <dbl>  
#> 1 Elf    Female      1229  
#> 2 Elf    Male        971  
#> 3 Hobbit Female        14  
#> 4 Hobbit Male      3644  
#> 5 Man    Female         0  
#> 6 Man    Male      1995  
#> 7 Elf    Female       183  
#> 8 Elf    Male       510
```

Filter for rows with `filter()`

Subset Observations (Rows)



Filter for rows with `filter()`

Keep only the rows with Elf characters

```
1 lotr %>%  
2   filter(race == "Elf")
```

```
#> # A tibble: 6 × 4  
#>   film                                race gender word_count  
#>   <chr>                             <chr> <chr>         <dbl>  
#> 1 The Fellowship Of The Ring      Elf   Female      1229  
#> 2 The Fellowship Of The Ring      Elf   Male         971  
#> 3 The Return Of The King          Elf   Female       183  
#> 4 The Return Of The King          Elf   Male        510  
#> 5 The Two Towers                  Elf   Female       331  
#> 6 The Two Towers                  Elf   Male        513
```

Filter for rows with `filter()`

Keep only the rows with Elf or Hobbit characters

```
1 lotr %>%  
2   filter((race == "Elf") | (race == "Hobbit"))
```

```
#> # A tibble: 12 × 4  
#>   film          race gender word_count  
#>   <chr>         <chr> <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Elf      Female      1229  
#> 2 The Fellowship Of The Ring Elf      Male        971  
#> 3 The Fellowship Of The Ring Hobbit    Female        14  
#> 4 The Fellowship Of The Ring Hobbit    Male     3644  
#> 5 The Return Of The King      Elf      Female      183  
#> 6 The Return Of The King      Elf      Male       510  
#> 7 The Return Of The King      Hobbit    Female         2  
#> 8 The Return Of The King      Hobbit    Male     2672
```

Filter for rows with `filter()`

Keep only the rows with Elf or Hobbit characters

```
1 lotr %>%  
2   filter(race %in% c("Elf", "Hobbit"))
```

```
#> # A tibble: 12 × 4  
#>   film          race gender word_count  
#>   <chr>         <chr> <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Elf      Female      1229  
#> 2 The Fellowship Of The Ring Elf      Male        971  
#> 3 The Fellowship Of The Ring Hobbit    Female       14  
#> 4 The Fellowship Of The Ring Hobbit    Male     3644  
#> 5 The Return Of The King      Elf      Female      183  
#> 6 The Return Of The King      Elf      Male       510  
#> 7 The Return Of The King      Hobbit    Female        2  
#> 8 The Return Of The King      Hobbit    Male     2672
```


Logic operators for `filter()`

Description	Example
Values greater than 1	<code>value > 1</code>
Values greater than or equal to 1	<code>value >= 1</code>
Values less than 1	<code>value < 1</code>
Values less than or equal to 1	<code>value <= 1</code>
Values equal to 1	<code>value == 1</code>
Values not equal to 1	<code>value != 1</code>
Values in the set <code>c(1, 4)</code>	<code>value %in% c(1, 4)</code>

Combine `filter()` and `select()`

Keep only the rows with Elf characters that spoke more than 1000 words, then select everything but the race column

```
1 lotr %>%  
2   filter((race == "Elf") & (word_count > 1000)) %>%  
3   select(-race)
```

```
#> # A tibble: 1 × 3  
#>   film                                gender word_count  
#>   <chr>                                <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Female      1229
```

Create new variables with `mutate()`

Make New Variables



Create new variables with `mutate()`

Create a new variable, `word1000` which is `TRUE` if the character spoke 1,000 or more words

```
1 lotr %>%  
2   mutate(word1000 = word_count >= 1000)
```

```
#> # A tibble: 18 × 5  
#>   film                                race  gender word_count word1000  
#>   <chr>                                <chr> <chr>      <dbl> <lgl>  
#> 1 The Fellowship Of The Ring Elf      Female    1229 TRUE  
#> 2 The Fellowship Of The Ring Elf      Male      971 FALSE  
#> 3 The Fellowship Of The Ring Hobbit Female     14 FALSE  
#> 4 The Fellowship Of The Ring Hobbit Male    3644 TRUE  
#> 5 The Fellowship Of The Ring Man      Female     0 FALSE  
#> 6 The Fellowship Of The Ring Man      Male    1995 TRUE  
#> 7 The Return Of The King Elf      Female    183 FALSE  
#> 8 The Return Of The King Elf      Male     510 FALSE
```

Handling if/else conditions

`ifelse(<condition>, <if TRUE>, <else>)`

```
1 lotr %>%
2   mutate(word1000 = ifelse(word_count >= 1000, TRUE, FALSE))
```

```
#> # A tibble: 18 × 5
#>   film                                race gender word_count word1000
#>   <chr>                                <chr> <chr>      <dbl> <lgl>
#> 1 The Fellowship Of The Ring Elf      Female    1229 TRUE
#> 2 The Fellowship Of The Ring Elf      Male      971 FALSE
#> 3 The Fellowship Of The Ring Hobbit Female     14 FALSE
#> 4 The Fellowship Of The Ring Hobbit Male    3644 TRUE
#> 5 The Fellowship Of The Ring Man      Female     0 FALSE
#> 6 The Fellowship Of The Ring Man      Male    1995 TRUE
#> 7 The Return Of The King Elf      Female    183 FALSE
#> 8 The Return Of The King Elf      Male    510 FALSE
```

Sort data frame with `arrange()`

Sort the `lotr` data frame by `word_count`

```
1 lotr %>%  
2   arrange(word_count)
```

```
#> # A tibble: 18 × 4  
#>   film                race  gender word_count  
#>   <chr>              <chr>  <chr>    <dbl>  
#> 1 The Fellowship Of The Ring Man    Female      0  
#> 2 The Two Towers      Hobbit Female      0  
#> 3 The Return Of The King Hobbit Female      2  
#> 4 The Fellowship Of The Ring Hobbit Female     14  
#> 5 The Return Of The King  Elf    Female    183  
#> 6 The Return Of The King  Man    Female    268  
#> 7 The Two Towers        Elf    Female    331  
#> 8 The Two Towers        Man    Female    401
```

Sort data frame with `arrange()`

Use the `desc()` function to sort in descending order

```
1 lotr %>%  
2   arrange(desc(word_count))
```

```
#> # A tibble: 18 × 4  
#>   film                race  gender word_count  
#>   <chr>              <chr>  <chr>      <dbl>  
#> 1 The Fellowship Of The Ring Hobbit Male      3644  
#> 2 The Two Towers          Man   Male      3589  
#> 3 The Return Of The King   Hobbit Male      2673  
#> 4 The Two Towers          Hobbit Male      2463  
#> 5 The Return Of The King   Man   Male      2459  
#> 6 The Fellowship Of The Ring Man   Male      1995  
#> 7 The Fellowship Of The Ring Elf    Female    1229  
#> 8 The Fellowship Of The Ring Elf    Male      971
```

Your turn

Read in the `data.csv` file in the "data" folder:

```
1 data <- read_csv(file.path("data", "data.csv"))
```

- Create a new data frame, `flights_fall`, that contains only flights that departed in the fall semester.
- Create a new data frame, `flights_dc`, that contains only flights that flew to DC airports (Reagan or Dulles).
- Create a new data frame, `flights_dc_carrier`, that contains only flights that flew to DC airports (Reagan or Dulles) and only the columns about the month and airline.
- How many unique airlines were flying to DC airports in July?
- Create a new variable, `speed`, in miles per hour using the `time` (minutes) and `distance` (miles) variables.
- Which flight flew the fastest?
- Remove rows that have `NA` for `air_time` and re-arrange the resulting data frame based on the longest air time and longest flight distance.

Week 1: *Getting Started*

1. Course Goal
2. Course Introduction
3. Quarto
4. Workflow & Reading In Data
5. Wrangling Data
6. Visualizing Data

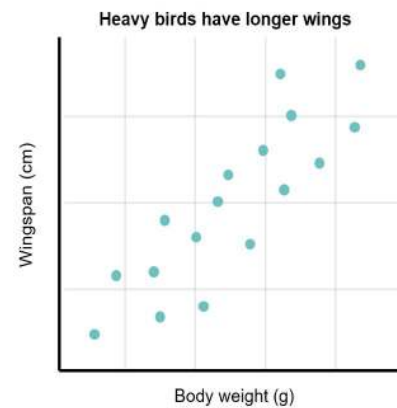
MAKING A GRAPH WITH GGPLOT2

Customise the look of your plot with themes
(pre-made or your own!):
`+ theme_bw()`

Add labels and titles:
`+ labs(x = "Body weight (g)", y = "Wingspan (cm)",
title = "Heavy birds have longer wings")`

Specify the type of graph and the variables to use:
`+ geom_point(aes(x = body.weight, y = wingspan))`

Plot the device containing your data:
`ggplot(data = birds)`



“Grammar of Graphics”

Concept developed by Leland Wilkinson
(1999)

ggplot2 package developed by Hadley
Wickham (2005)

Making plot layers with ggplot2

1. The data
2. The aesthetic mapping (what goes on the axes?)
3. The geometries (points? bars? etc.)
4. The annotations / labels
5. The theme

Layer 1: The data

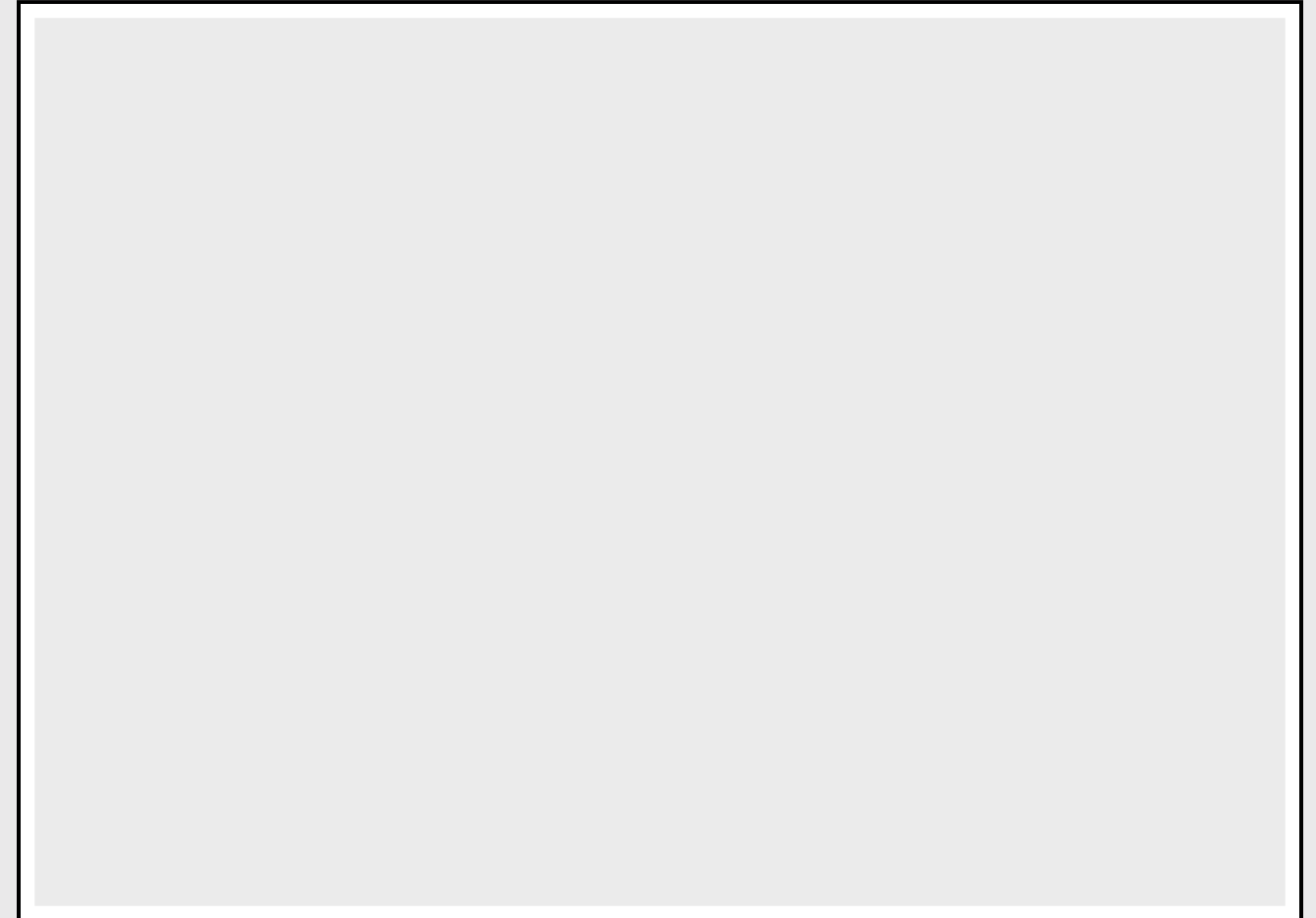
```
1 head(mpg)
```

```
#> # A tibble: 6 × 11
#>   manufacturer model displ  year   cyl trans      drv    cty   hwy fl      class
#>   <chr>         <chr> <dbl> <int> <int> <chr>    <chr> <int> <int> <chr> <chr>
#> 1 audi         a4      1.8  1999     4 auto(l5)  f      18    29 p      compact
#> 2 audi         a4      1.8  1999     4 manual(m5) f      21    29 p      compact
#> 3 audi         a4      2    2008     4 manual(m6) f      20    31 p      compact
#> 4 audi         a4      2    2008     4 auto(av)   f      21    30 p      compact
#> 5 audi         a4      2.8  1999     6 auto(l5)  f      16    26 p      compact
#> 6 audi         a4      2.8  1999     6 manual(m5) f      18    26 p      compact
```

Layer 1: The data

The `ggplot()` function initializes the plot with whatever data you're using

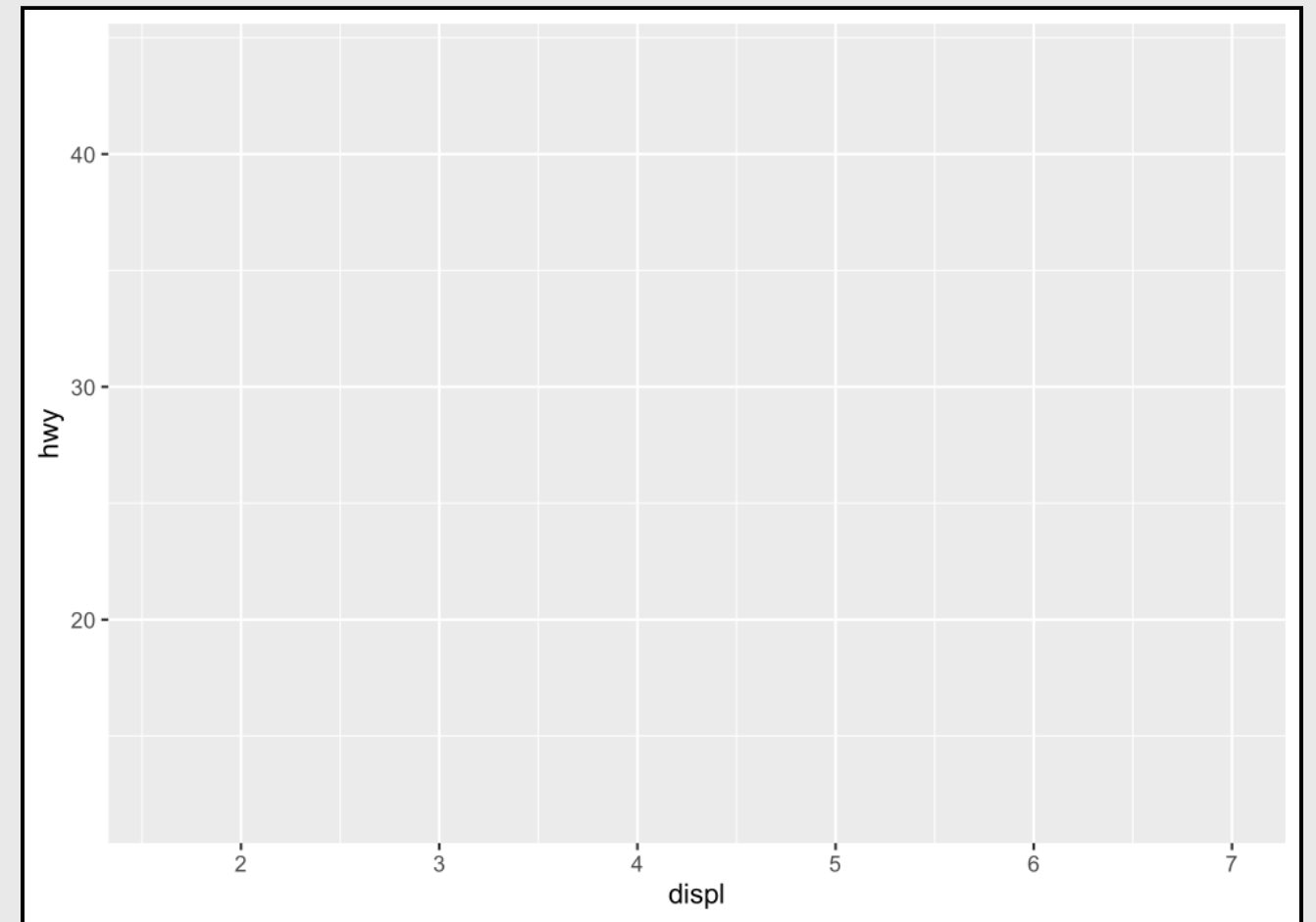
```
1 mpg %>%  
2   ggplot()
```



Layer 2: The aesthetic mapping

The `aes()` function determines which variables will be *mapped* to the geometries (e.g. the axes)

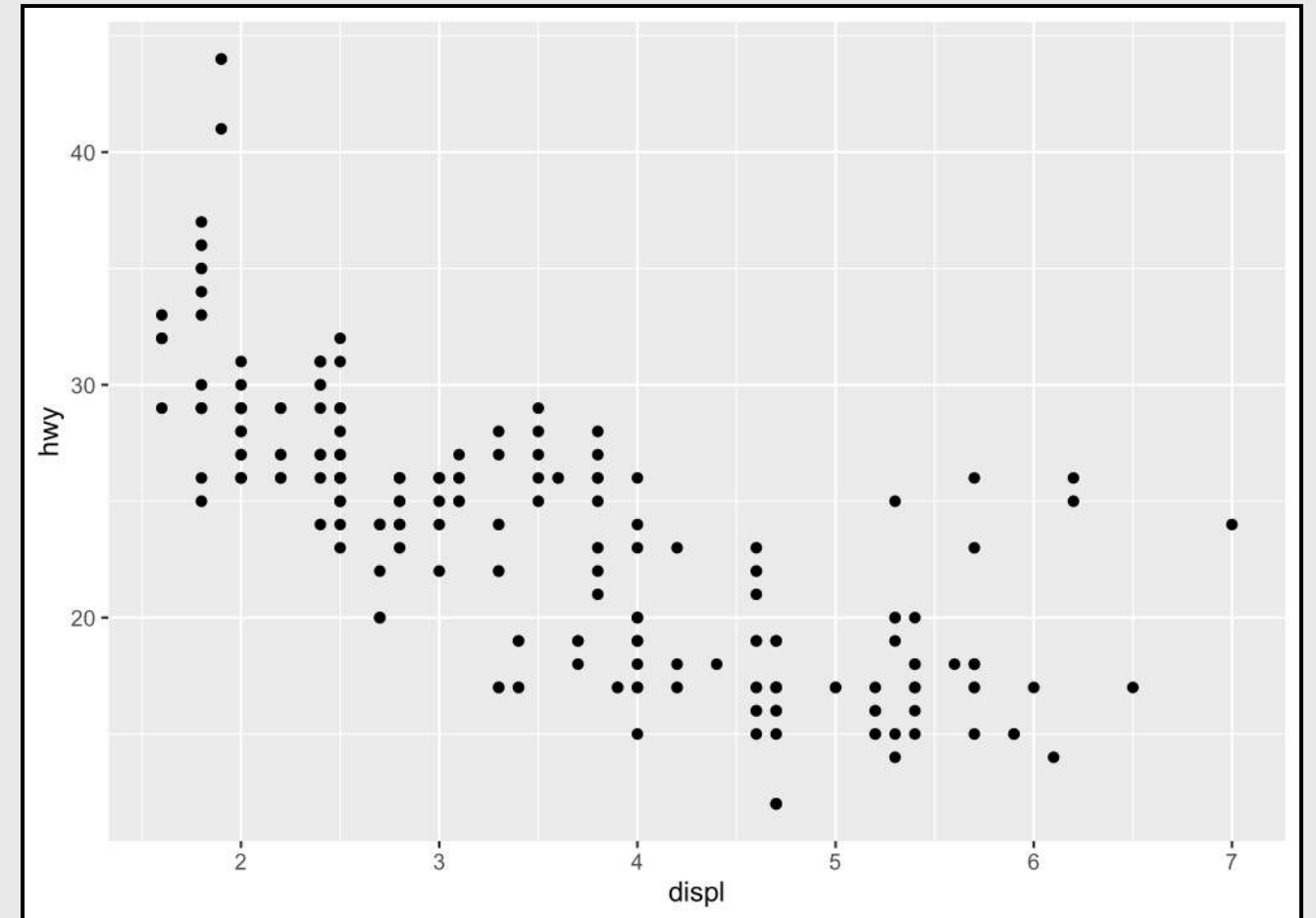
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy))
```



Layer 3: The geometries

Use `+` to add geometries, e.g. `geom_points()` for points

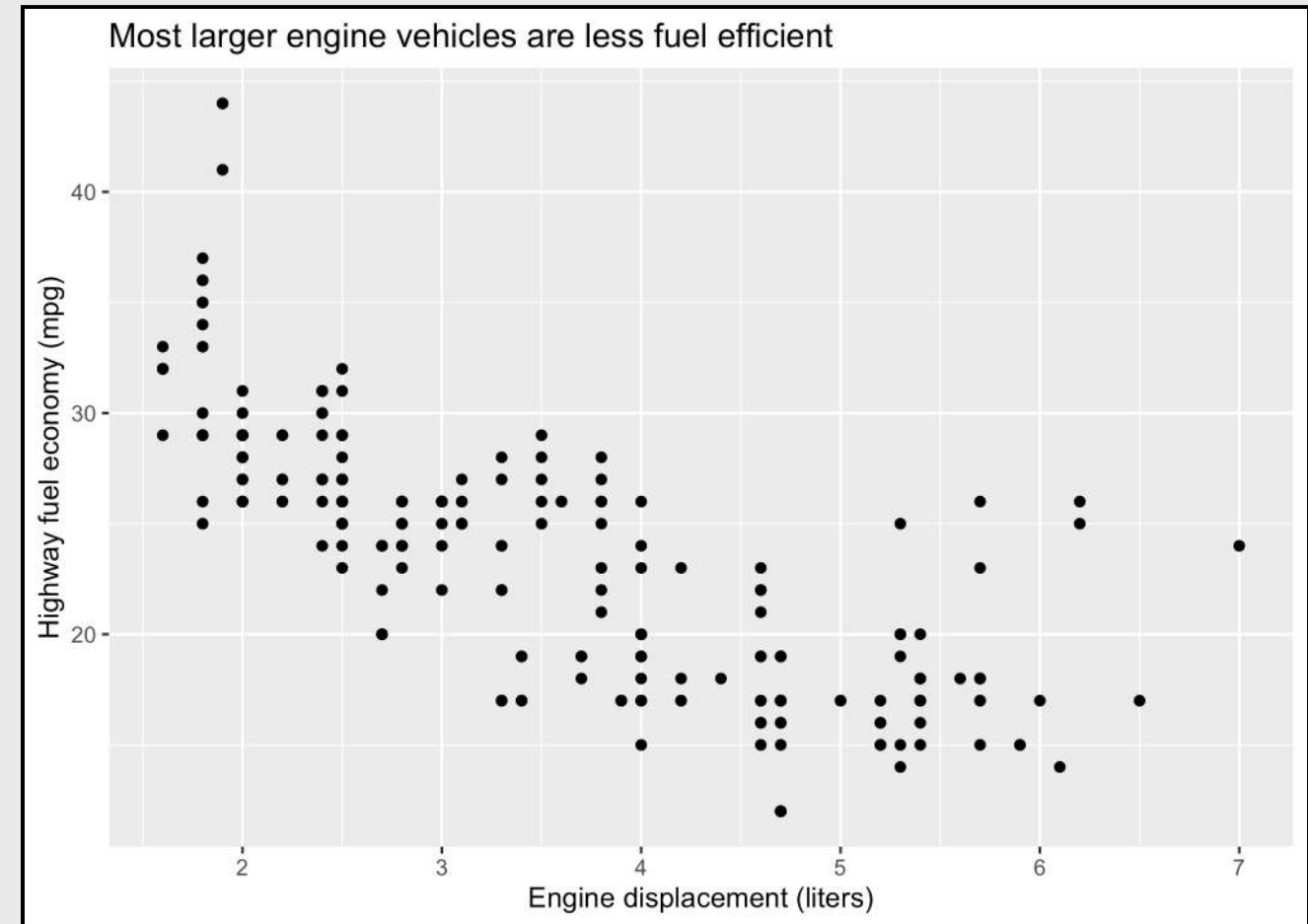
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point()
```



Layer 4: The annotations / labels

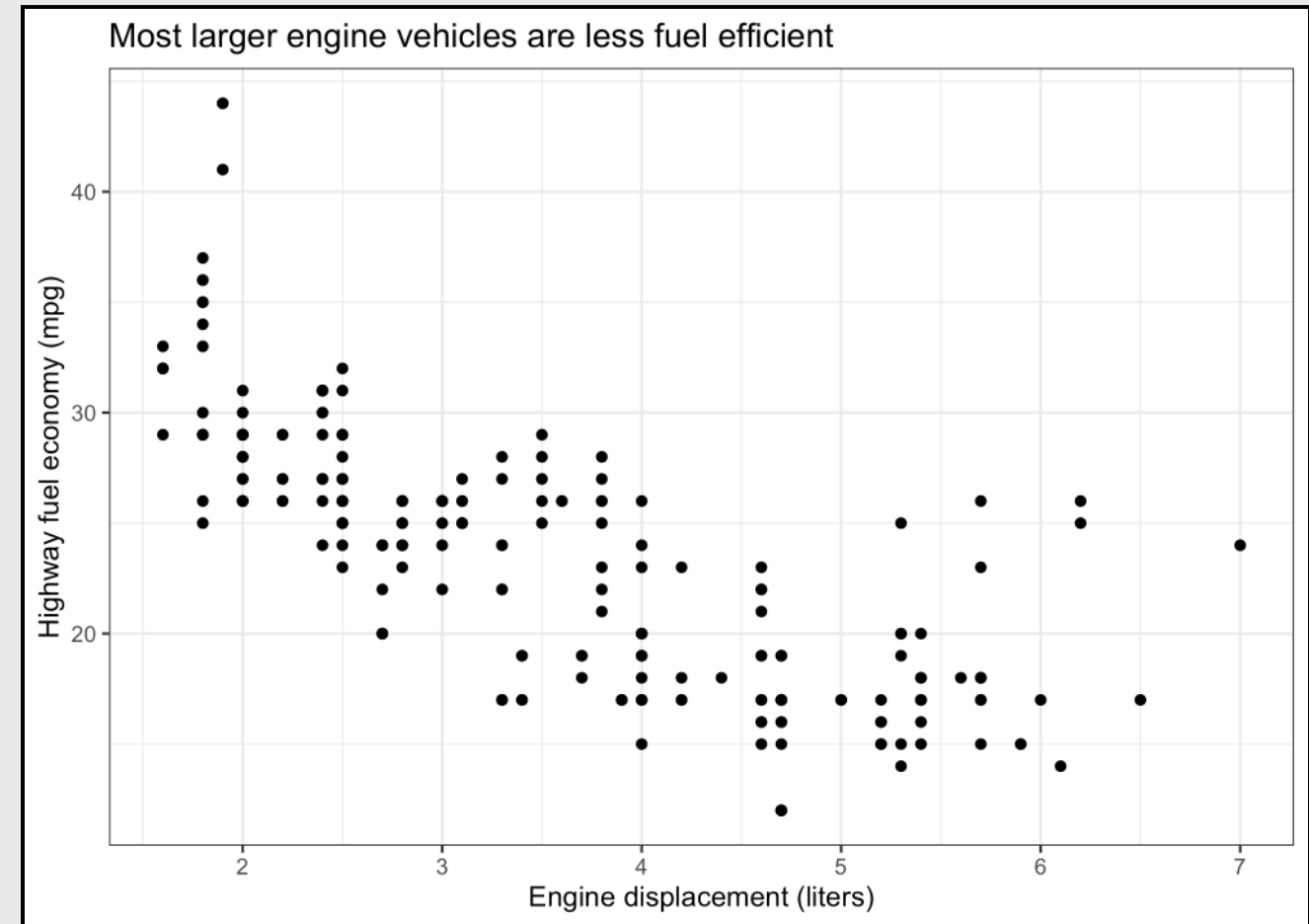
Use `labs()` to modify most labels

```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   labs(  
5     x = "Engine displacement (liter  
6     y = "Highway fuel economy (mpg)  
7     title = "Most larger engine veh  
8   )
```



Layer 5: The theme

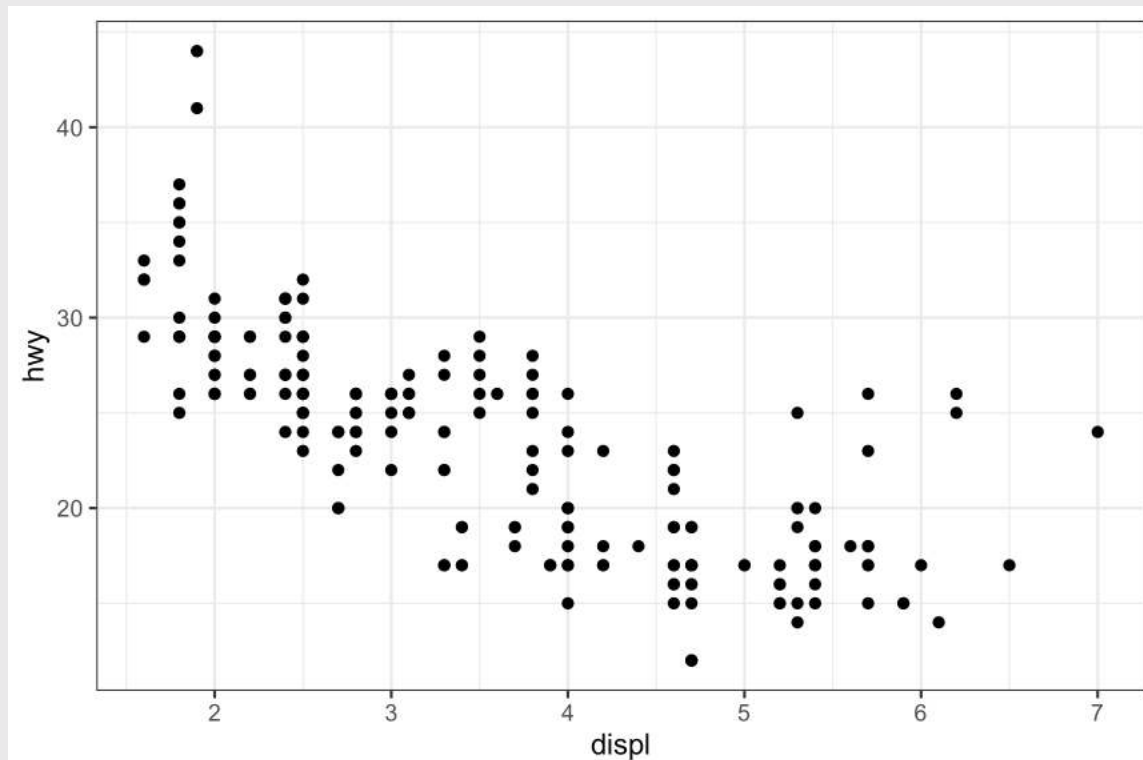
```
1 mpg %>%
2   ggplot(aes(x = displ, y = hwy)) +
3   geom_point() +
4   labs(
5     x = "Engine displacement (liter
6     y = "Highway fuel economy (mpg)
7     title = "Most larger engine veh
8   ) +
9   theme_bw()
```



Common themes

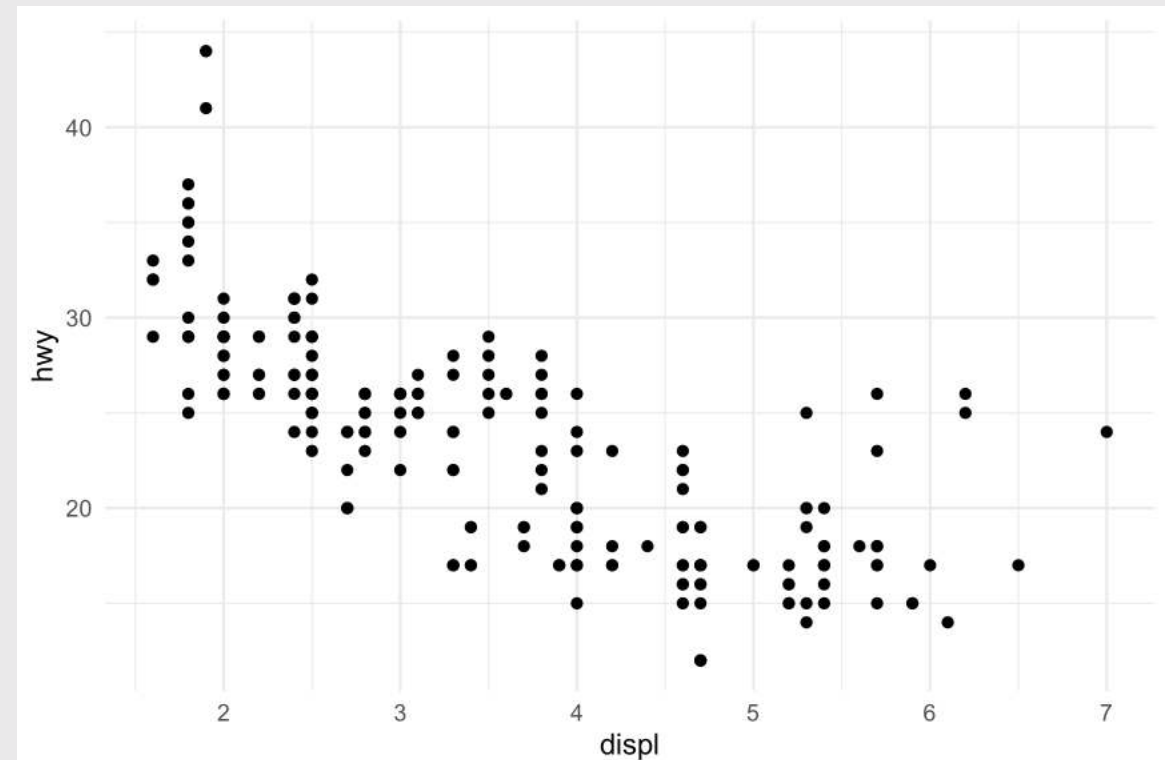
theme_bw()

```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   theme_bw()
```



theme_minimal()

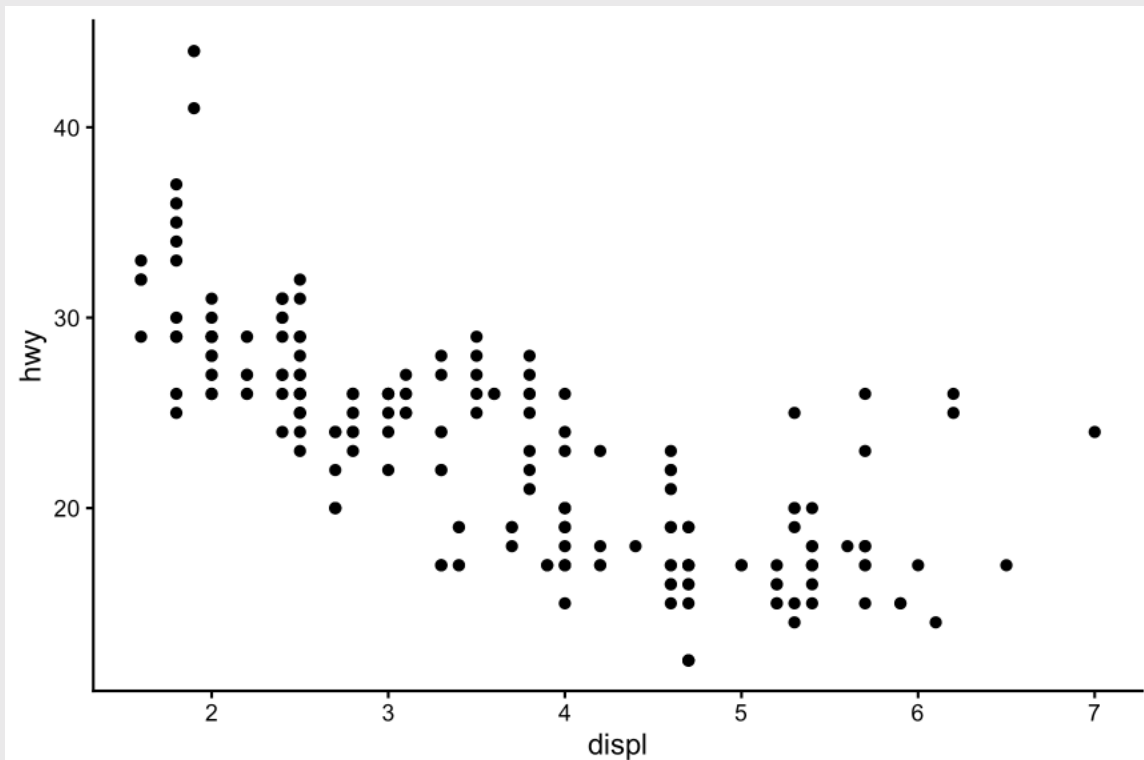
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   theme_minimal()
```



Common themes

theme_classic()

```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   theme_classic()
```



theme_void()

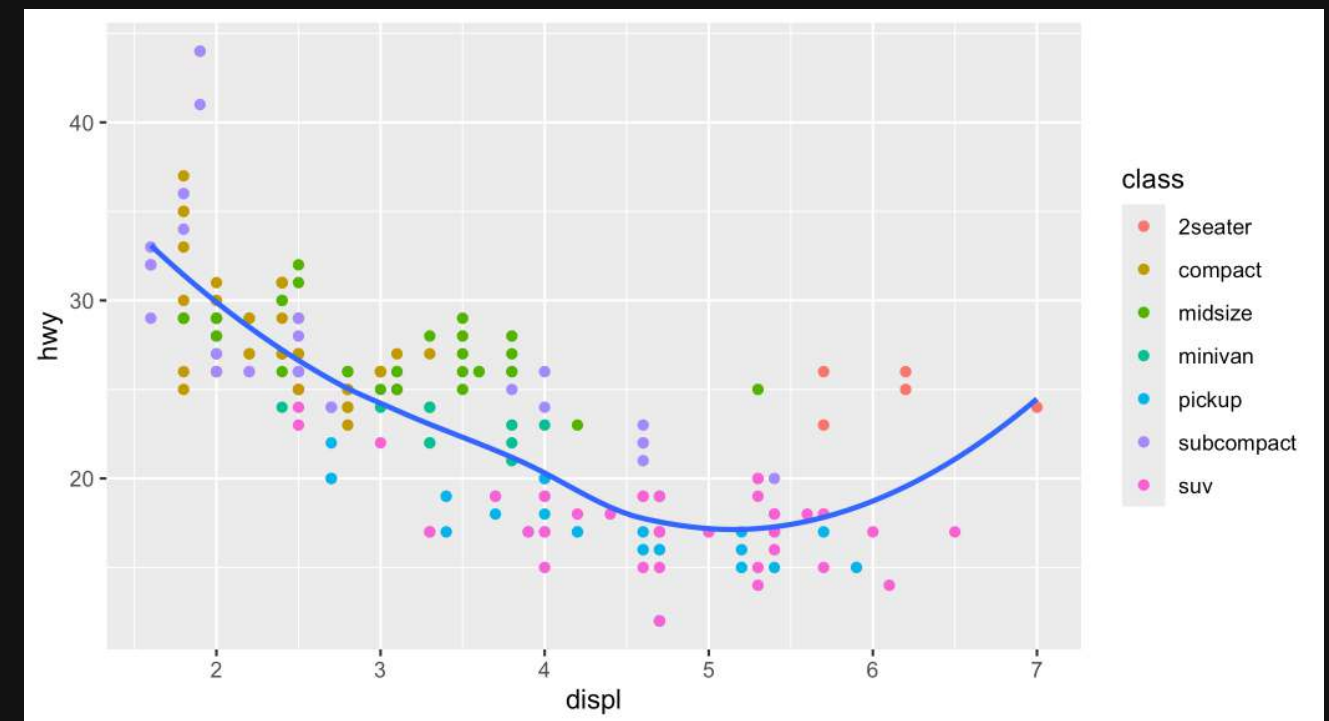
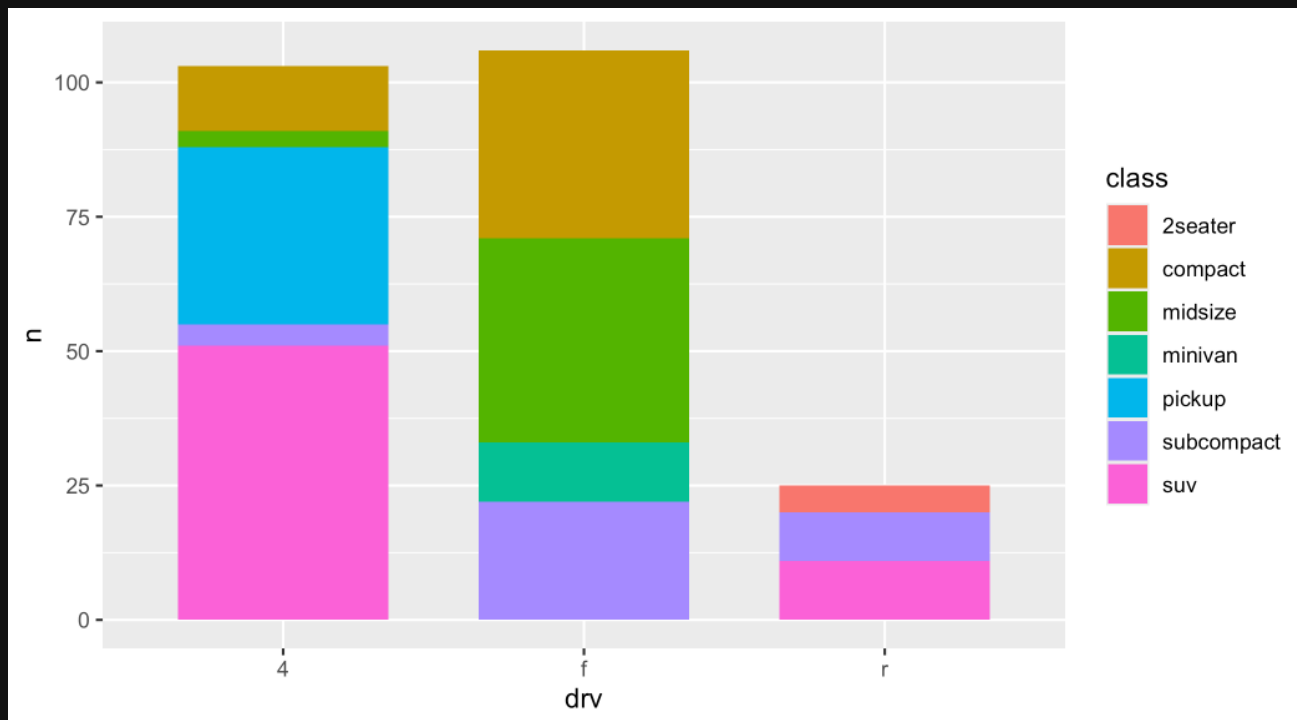
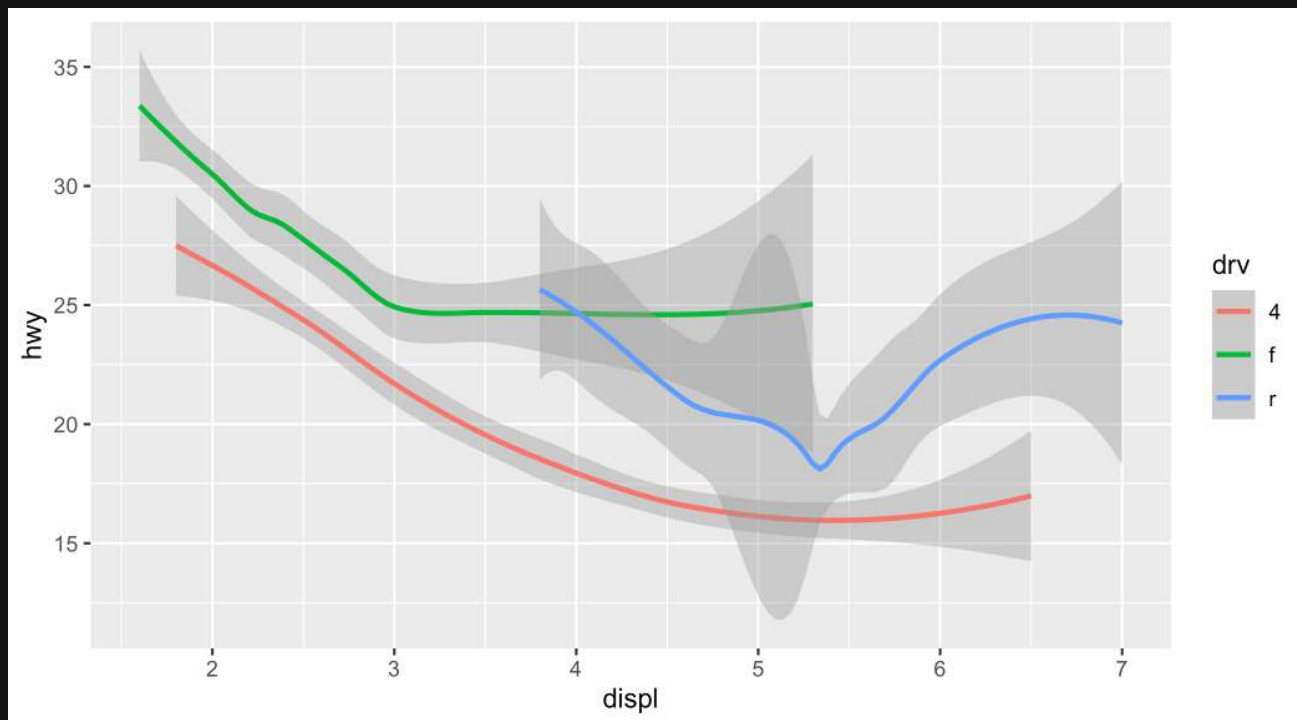
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   theme_void()
```



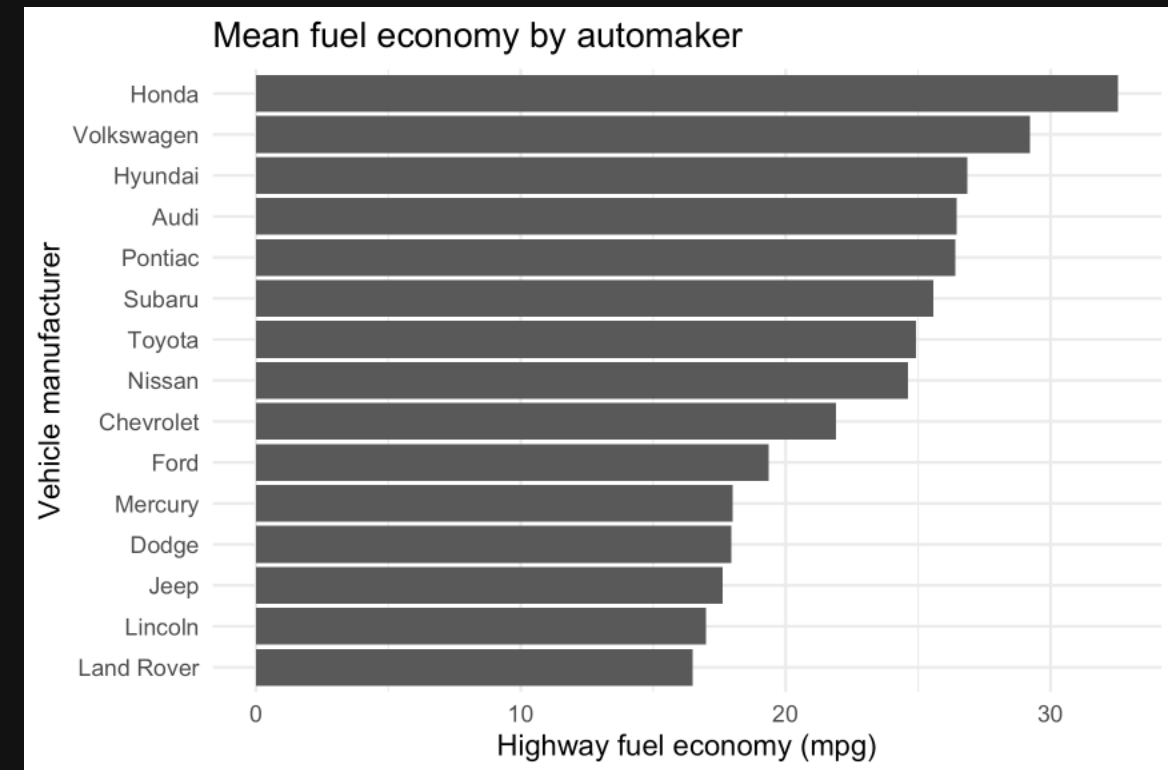
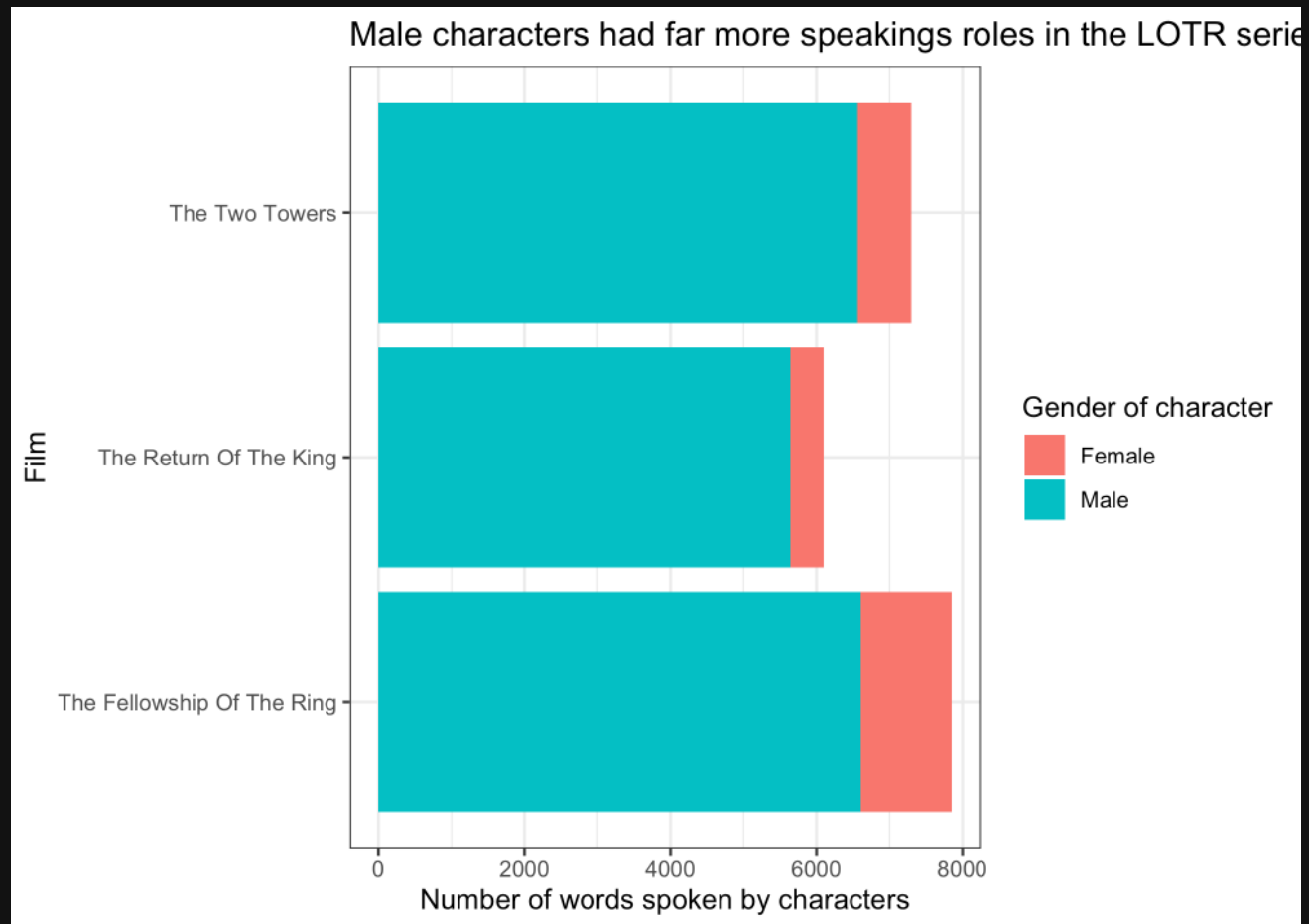
Your turn

Open `practice.qmd`

Use the `mpg` data frame and ggplot to create these charts



Extra practice



Use [this link](#) to form teams